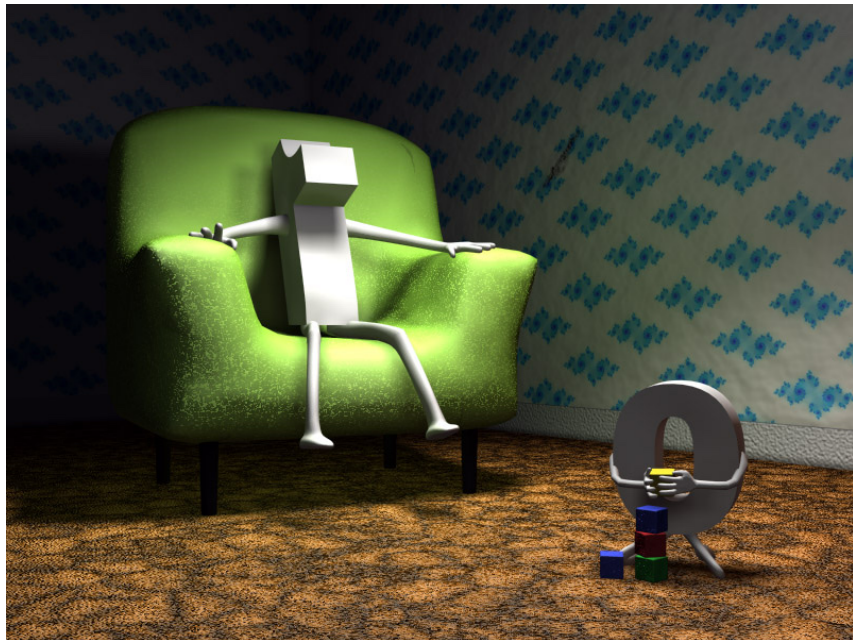


Chapter 1. INTEGERS

1. 1. Induction



In this short video we will talk about **STRONG INDUCTION**.

STRONG INDUCTION. Let $m \in \mathbb{N}$ and for each $n \geq m$, let p_n be a statement. If the following conditions are satisfied:

- (1) p_m is true.
- (2) If $k \geq m$ and all of $p_m, p_{m+1}, \dots, p_k$ are true, then p_{k+1} is also true.

Then p_n is true for every $n \geq m$.

Proof. For $n \geq m$, we define the statement

t_n : $p_m, p_{m+1}, \dots, p_n$ are true.

Basis step. t_m is true by (1).

Inductive step. Suppose that t_k is true for some $k \geq m$ and let us show that t_{k+1} is true.

Since t_k is true, we have that $p_m, p_{m+1}, \dots, p_k$ are true, and (2) tells us that p_{k+1} is true. This shows that t_{k+1} is true. By the Principle Mathematical Induction, we have that p_n is true for all $n \geq m$. ■

Example 1.4. Use strong induction to show that every integer $n \geq 2$ is a product of (one or more) primes.

Solution. For $n \geq 2$, we define

P_n : n is a product of primes.

(1) P_2 is true because 2 is a prime.

(2) Take $K \geq 2$ and assume that

$P_2, P_3, \dots, P_K$ are true. We need to show that P_{K+1} is true. That is, we need to prove that $K+1$ is a product of primes.

We consider two cases:

CASE 1. $K+1$ is a prime.

There is nothing to do in this case.

CASE 2. $k+1$ is NOT a prime.

We can write $k+1 = ab$, where
 $2 \leq a, b \leq k$. Since we are assuming
that $P_2, \dots, P_k$ are true, we have
that a and b are both product
of primes, and so their product, i.e., $\underbrace{ab}_{k+1}$.

This shows that P_{k+1} is true, and
by Strong Induction P_n is true for
all $n \geq 2$.

